MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8561

This specification is approved for use by the Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
Outline: See Figure 1
Mounting Position: Any
Weight: 4 ounces nominal

ABSOLUTE RATINGS: F1 = 110 MHz

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ehk	Pg1	Pg2
Unit:	Vac (Note 1)	Vdc	Vdc	Vdc	mAdc	Vdc	W	W
Maximum:								
C Teleg:	6.0 ±5%	2,000	-250	300	400	±150	1.0	8.0
C Telep:	6.0 ±5%	1,500	-250	300	300	±150	1.0	8.0
AB or A:	6.0 ±5%	2,000	----	400	400	±150	1.0	8.0
TEST CONDITIONS:	6.0	1,000	Adj.	300	240	----	---	---

ABSOLUTE RATINGS:

Parameter:	Pp	tk	T(Seals & Anode Core)	Cooling	Barometric pressure, reduced
Unit:	W	sec (min)	°C		mmHg
Maximum:				(Note 2)	
C Teleg:	400	30	250	---	141
C Telep:	250 (Note 3)	30	250	---	141
AB or A:	400	30	250	---	141
TEST CONDITIONS:	---	120	---	(Note 4)	---

GENERAL:

Qualification - Required

Ⓓ Denotes changes

8561

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1261	Electrode voltage (grid)		0.65	II	Ec1	-32.0	-45.0	Vdc
1266	Total grid current	Eb = 2,000 Vdc; Ib = 150 mAdc See note 5	0.65	II	Ic1	---	-15	µAdc
1256	Electrode current (screen)		0.65	II	Ic2	-7.0	+3.0	mAdc
1266	Primary grid emission (control)	Pg1 = 1 watt; t = 15; anode and screen grid floating	0.65	II	Isg1	---	-25	µAdc
1266	Primary grid emission (screen)	Ec1 = 0; t = 15; Pg2 = 8 watts; anode floating	0.65	II	Isg2	---	-250	µAdc

2212	Pulse emission	Eb = Ec2 = 250 Vdc;
		Ec1 = -100 Vdc;
		egk/ik = 2.4 a;
		prr = 11 +1;
		tp = 4,500 µs (min);
		tr = tf = 25 µs;
		slope = 0.5%
		ripple = 0.1%
		Ef = 5.4 Vac
		See note 6
			0.65	II	Δik	---	200	ma

1372	Current division (Method A)	Eb = Ec2 = 250 Vdc;
		Ec1 = -100 Vdc;
		egk/ib = 1.6 a;
		prr = 11 +1;
		tp = 4,500 µs (min)
			0.65	II	egk
					ic1
					ic2
						8.0
						--- 	
						--- 	
							18.0
							300
							320
								V
								ma
								ma

	Quality conformance inspection, part 2							
1301	Heater current		----	--	If	3.15	4.00	Aac
1331	Direct-interelectrode capacitance (grounded cathode connection)		----	--	Cgp Cin Cout	---- 30.0 3.9	0.07 38.0 5.0	pF pF pF
1336	Heater-cathode leakage	Ehk = +250 Vdc Ehk = -250 Vdc	----	--	Ihk Ihk	---- ----	150 150	µAdc µAdc
	Quality conformance inspection, part 3	See note 7						
----	Vibration (noise)	Ef = 6.0 (ac or dc) Ebb = 2,500 Vdc; Ec2 = 400 Vdc; Rp = 4,900 ohms; Ec1/Ib = 100 mAdc; Accel= 10G peak(min); F = 28 to 500 Hz, ascending only; X and Y axes only See note 8	----	--	Ep	----	30	Vac
----	Vibration (noise) end points:							
1261	Electrode voltage (grid)		----	--	Ec1	-32.0	-45.0	Vdc
1266	Total grid current		----	--	Ic1	----	-20	µAdc
1042	Shock, specified pulse, Condition A	Eb = 2,000 Vdc; Ec2 = 350 Vdc; Ec1 = -150 Vdc See note 9	----	--	---	----	----	----

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued	See note 7						
	Shock, specified pulse end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-32.0	-45.0	Vdc
1266	Total grid current		---	---	Ic1	-----	-20	µAdc
----	Life-test (1) provisions	Group C; Ef = 6.6 Vac; Ec1 = Ec2 = Eb = 0; t = 500 hours	---	---	---	-----	-----	----
----	Life-test (1) end points:							
1366	Interelement leakage resistance, cold	Rs = 2.5 Meg E = 100 Vdc (g1 neg) E = 500 Vdc (g2 pos) See note 10	--- ---	--- ---	Rg1k Rg1g2	100 100	----- -----	Meg Meg
----	Life-test (3) provisions	Group C; Heater cycling; 120 sec(approx.) on 240 sec(approx.) off g2 and anode floating t = 200 hrs See note 11	---	---	---	-----	-----	---
---	Life-test (3) end points:							
1301	Heater current		---	---	If	3.15	4.00	Aac
Ⓓ 1136	Heater cathode leakage	Ehk = +250 Vdc Ehk = -250 Vdc	--- ---	--- ---	Ihk Ihk	----- -----	150 150	µAdc µAdc
----	Cooling and pressure drop	Eb = 2,000 Vdc; Ec1/Ib = 200 mAdc; Air flow = 7.4 cfm; See notes 12 and 14	---	---	Press: T anode core	----- -----	0.8 250	$In.H_2O$ °C
----	Life-test (2) provisions	Ebb = 1,000 Vdc; Ec2 = 250 Vdc; Ec1/Ib = 50 mAdc; Rp = 1,000 ohms TA = 200°C minimum; Eg1/Ib = 100 mAdc; F = 1 kHz minimum; t = 100 hrs See notes 13 and 14	---	---	---	-----	-----	---
----	Life-test (2) end points:							
2212	Pulse emission	Except Ef = 6.0 Vac	---	---	δik	-----	100	ma
1266	Total grid current		---	---	Ic1	-----	-20	µAdc
1002	Barometric pressure reduced	E = 2,500 Vac p to g2; no other voltages applied, no socket; pressure = 141 ±5 mmHg absolute (max)	---	---	t	30	-----	Sec

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NOTES:

Ⓓ 1. When long life and consistent performance are factors, the applied heater voltage should be maintained within plus or minus five percent of nominal. Heater voltage should be measured directly at the tube or socket terminals with an accurate meter, preferably an rms-responding type.

2. When the tube is operated at 100 percent of the maximum rated anode dissipation at an incoming air temperature of 25°C maximum, a minimum air flow of 7.4 cfm at sea level shall pass through the anode cooler. If the EIMAC socket SK-700, or equivalent, is used, an incoming air flow of 7.4 cfm to the lower end of the socket is required. An air director (chimney) such as the EIMAC SK-606, or equivalent, should also be used. At this flow of 7.4 cfm, the static pressure drop across the tube/socket/chimney combination is approximately 0.7 inch of water. This pressure drop varies with the amount of escaping air and with the shape and construction of the air director. Air cooling of the tube should be increased with increased incoming air temperature, higher altitude operation, or a combination. In all cases of operation, a socket which provides forced-air cooling of the base shall be used. Air flow must be applied before or simultaneously with the application of electrode voltages (including heater) and may be removed simultaneously with them. In all cases of operation, sufficient cooling must be provided to prevent seal and anode core temperatures in excess of the specified maximum value, and in cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

3. Applies to carrier-only conditions.

Ⓓ 4. In all electrical tests involving application of heater voltage, forced-air cooling of the tube is allowable at the rate of 5.0 cfm maximum for the base and anode, unless otherwise specified. A separate source may be used for the base and anode but neither shall exceed 5.0 cfm. The tube may be operated in an air-system socket (EIMAC SK-700 family, with SK-606 chimney, or equivalent). The cooling air shall not have a temperature of less than 20°C nor an absolute pressure greater than 32 inches of mercury. If testing is performed at an altitude significantly above sea level, air flow may be corrected for comparable mass flow rate.

5. This test is to be the first test performed at the conclusion of the holding period.

Ⓓ 6. For the basic test circuit, see Test Method 1372, figure 1372-1.

Ⓓ 7. The Vibration and Shock Tests listed under Quality Conformance Inspection, Part 3 shall be performed quarterly (every 3 months), with sample sizes and failure allowances as follows:

Sample Size	Failure Allowance
$N_1 = 4$	$C_1 = 0$
$N_2 = 4$	$C_2 = 1$

These tests shall not be considered destructive except in case of failure. Tubes meeting the post-test end points shall be considered to be in acceptable physical condition. In case of a sample failure, that test shall become Quality Conformance Inspection, Part 2, AQL 6.5, Inspection Level S3, for three consecutive successful submissions, at which time the test will revert to the quarterly basis.

8. The fixtures described in Drawing 284-JAN shall be used to hold the tube during the test. The test circuit shown in Figure 3 shall be used, but tubes found to electrically oscillate for causes other than vibration shall not be tested nor rejected on this test. Each tube under test shall be subjected to one sweep cycle in each of the two axes X and Y. One sweep cycle (28 to 500 Hz, ascending only) shall be covered in 6 to 12 minutes. Each tube shall be vibrated for 60 seconds at the frequency which gives the maximum vibration output voltage in each of the two axes. If at the end of 60 seconds the vibration output is still increasing, the vibration shall be continued until there is no further increase for 60 seconds.

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NOTES: -Continued

8. Cont'd The tubes shall not show noise voltage in excess of the maximum limit specified, except one intermittent short per tube shall be allowable during this test. In addition to reading noise voltage output with the VTVM, such as the HP 400 D, or equivalent, a permanent recording shall be made using a good quality recorder to produce a plot of noise voltage versus frequency. Noise voltage amplifiers used with the recorder shall have a +dB frequency response over the range to be measured (28 - 500 Hz) and the overall recording equipment shall be capable of fast response in order to show sharp noise voltage spikes resulting from internal tube resonances or other phenomenon. Prominent noise peaks indicated on the recording shall be individually investigated by fixed-frequency operation, and the 60-second operation specified above shall be made at the frequency of highest noise as so selected. The frequency at the extremes of the sweep shall be read with an accuracy of ±1 percent above 100 Hz and ±1 Hz below 100 Hz.

9. The fixture described in Drawing 284-JAN, shall be used to hold the tube under test. The tubes shall be subjected to the specified acceleration in X, Y, and Z axes, with 6 shocks in each axis. Tubes showing any permanent shorts, or more than one temporary short during this test, shall be rejected.

10. This test shall be made at least 30 minutes after Ef is turned off. Rated air flow shall be maintained during the 30-minute interval. Use General Radio Megohmeter Model 1862C, or equivalent. Tube elements not under test should be left floating.

11. No grid-cathode shorts are permitted during or after life test.

12. An infinite baffle system as shown in Figure 2, or equivalent, shall be used. The tube shall be mounted in an air-system socket (EIMAC SK-700, SK-710, or equivalent, with an SK-606 or equivalent chimney), so arranged that the entire air flow passes through the anode cooler, in a base-to-anode direction. If the test is not performed at sea level, an equivalent mass flow shall be used. Static pressure drop is measured across the tube/socket/chimney combination. Temperature shall be measured by means of a thermocouple, located as follows: the thermocouple shall be embedded in the top of the anode core, adjacent to the cooler fins, by means of drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple firmly in place. Good electrical continuity between the thermocouple and the metal area in close proximity shall be demonstrated before the cooling part of the test is performed.

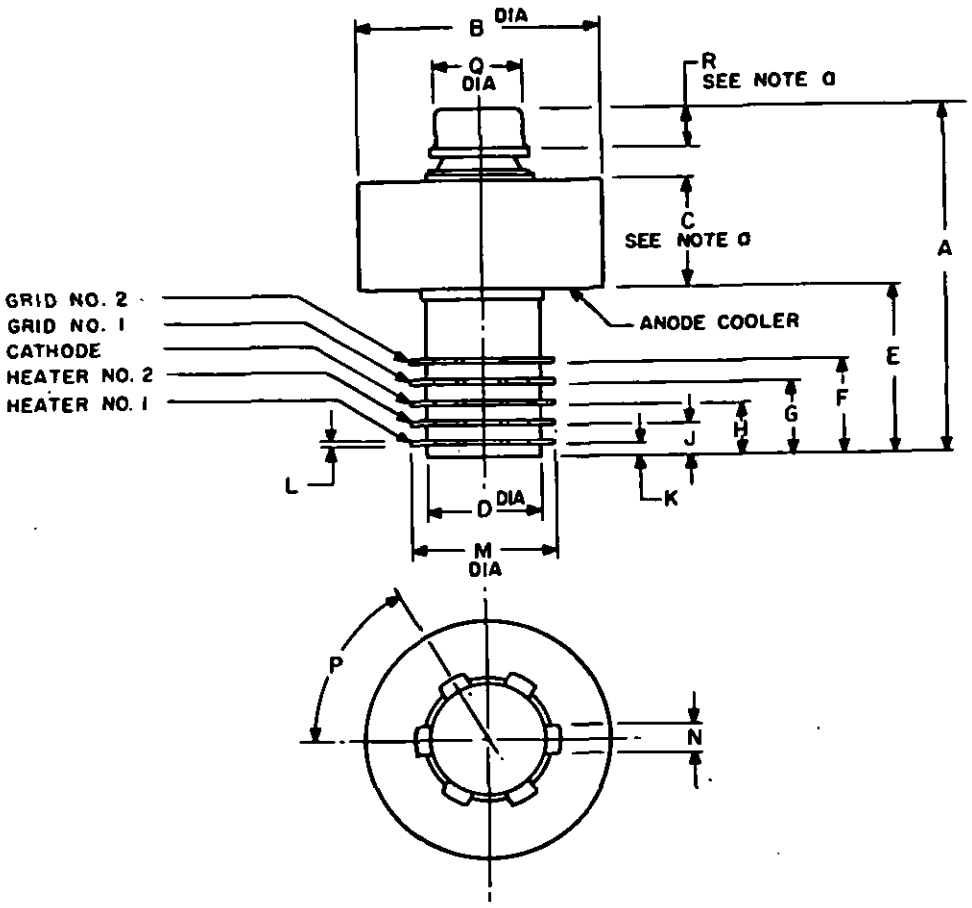
13. Any suitable high-temperature socket may be used. Note 14 shall apply except the quantity tested shall be one.

14. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test shall be made a part of Quality Conformance Inspection, Part 2, level S3, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

(D) 15. Reclaimed materials shall be utilized to the maximum extent possible as cited in MIL-STD-961 and MIL-STD-962 within the quality limits required by this document and to fulfill compliance with the Resource/Recovery Act of 1976 (Public Law 94-580 dated 21 Oct 76).

(D) 16. Changes from Previous Issue The margins of this revision are marked with a letter D to indicate where changes from the previous revision have been made. This was done as a convenience only and the Government assumes no liability for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this revision without regard to the marginal notations and the relationship to the previous issue of this document.

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DIMENSIONS								
DIM	INCHES		MILLIMETERS		DIM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 2					QUALITY CONFORMANCE INSPECTION, PART 3 (see note b)				
A	2.300	2.500	58.42	63.50	B	1.610	1.640	40.89	41.66
D	.740	.770	18.80	19.56	C	.710	.790	18.03	20.07
F	.602	.642	15.29	16.31	E	1.133	1.195	28.78	30.35
G	.470	.500	11.94	12.70	N	.170	.185	4.32	4.70
H	.329	.359	8.36	9.12	Q	.559	.573	14.20	14.55
J	.193	.213	4.90	5.41	R	.240		6.10	
K	.050	.072	1.27	1.83	REFERENCE DIMENSIONS				
L	.010	.020	.25	.51	P	60°			
M	.936	.956	23.77	24.28					

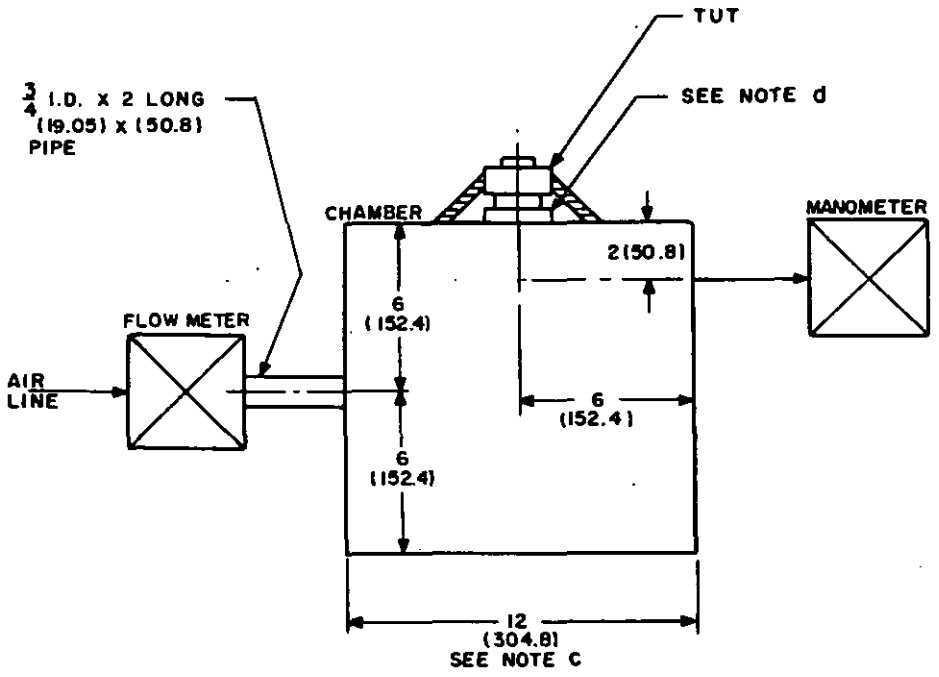
NOTES:

a. Available anode contact surfaces.

Ⓓ b. See note 14.

FIGURE 1 Outline drawing of electron tube type 8561

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NOTES:

a. All dimensions are in inches.

b. Metric equivalents in parentheses are shown for information only and are based upon 1 inch = 25.4 mm.

c. 12 inch (304.8) cube inside dimensions (or equivalent) shall be free of air leaks.

d. Socket shall be EIMAC, SK-700, SK-710 or equivalent.

FIGURE 2. Baffle system

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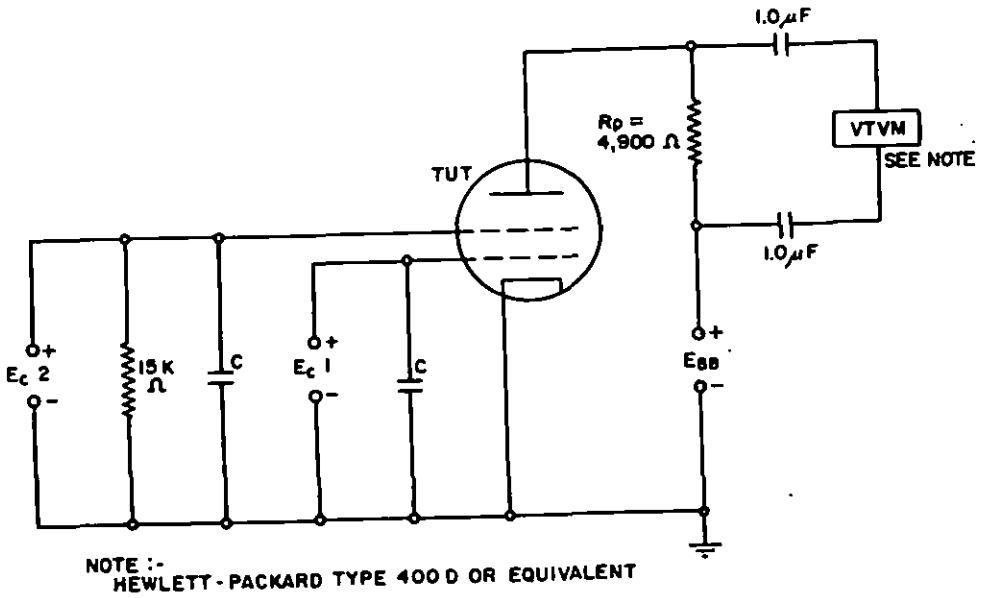


FIGURE 3 Basic test circuit for vibration (noise) test

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MIL-E-1/1541D Electron Tube, Power Type 8561

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WPAFB, OH 45433

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